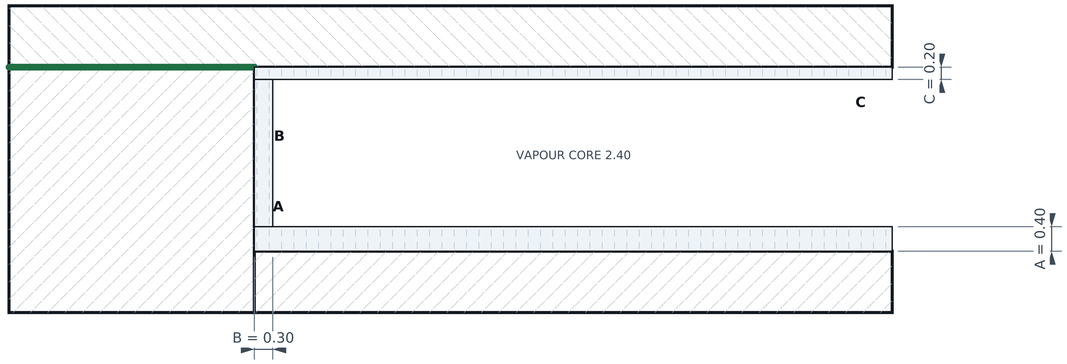
WICK ZONE SECTION

SCALE 50:1 — ASSEMBLED



Zones A, B and C are sintered in ONE cycle and must be capillary-continuous:

condensate returns C → B → A. A break at the B/A junction causes evaporator dry-out.

WICK ZONES

ZONE	LOCATION	THK	FUNCTION
A	Pocket floor, around posts	0.40	evaporator + liquid supply
B	Pocket side walls (4 off)	0.30	perimeter return path
C	Lid underside, posts masked	0.20	condensate collection

SINTER PROCESS

PARAMETER	VALUE	TOL / NOTE
Cu powder, spherical, gas-atomised	**45 – 75 um**	sieved, -200/+325 mesh
Effective pore radius	**12.6 um**	= 0.21 x d(particle)
Porosity	**55 %**	± 5 %, quantitative metallography
Permeability (Blake-Kozeny)	1.97e-11 m2	derived, not measured
Capillary head, water @ 60 C	**10.5 kPa**	vs 0.66 kPa gravity head
Sintering temperature	950 C	± 10 C
Dwell	30 min	at temperature
Atmosphere	H2 (dew pt < -40 C)	or vacuum < 1e-3 Pa
Ramp / cool	10 C/min / furnace cool	no quench
Tooling	graphite mandrel + mask	masks post tops, rim, fill port

WICK ACCEPTANCE

ACCEPTANCE TEST	REQUIREMENT	METHOD
Porosity	55 % ± 5	image analysis, 3 sections
Wick thickness A	0.40 ± 0.05	cross-section, 5 points
Capillary rise, water	> 25 mm in 60 s	witness coupon, vertical
Groove continuity	open, no blockage	back-light / CT, first article

POWDER TRADE STUDY — WHY 45-75 um

POWDER	r eff	dP cap	Q cap VERT	VERDICT
25 um	5.2 um	25.2 kPa	348 W	clogs, low K
45 – 75 um	**12.6 um**	**10.5 kPa**	**804 W**	**SELECTED**
100 um	21.0 um	6.3 kPa	1279 W	coarse, ok
150 um	31.5 um	4.2 kPa	1802 W	marginal tilt
400 um (original drawing)	100 um	0.4 kPa	0 W	FAILS — < gravity

The original 200 um pore spec gives 380 Pa of capillary head against a 500 Pa gravity head

over 68 mm: the wick cannot lift liquid at all in any vertical orientation.

CAPILLARY PRESSURE BUDGET — WORST CASE, VERTICAL

TERM	VALUE	NOTE
Capillary head available 2σ/r	**10 508 Pa**	water @ 60 C, r = 12.6 um
Liquid loss through wick + grooves	− 12.6 Pa/W	at 804 W: 10 127 Pa
Vapour loss through 2.40 core	− 0.2 Pa/W	negligible
Gravity head, 68 mm adverse	− 656 Pa	vertical, evaporator up
Margin at 113 W design point	**+ 8 428 Pa**	**7.4 x**

The wick is sized by the vertical case. Horizontal operation carries 858 W and is never the limit.

PROCESS NOTES

1. Mask post top faces and the 4.0 rim before sintering — sintered powder on a bond land destroys hermeticity.
2. Mask the fill port bore. Powder in the pinch-off tube prevents a gas-tight cold weld.
3. Grooves must remain open under the wick. Verify by back-light or CT on the first article.
4. Sinter and diffusion bond may share one furnace cycle only if the bond fixture applies load after the sinter dwell; otherwise run separately.
5. No organic binders. Any residue outgasses into the sealed chamber as non-condensable gas.

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **SINTERED WICK — PROCESS SPECIFICATION** (IN-SITU SINTERED Cu POWDER / ALL WICK ZONES)	SCALE: **50:1** (THIRD ANGLE PROJECTION)	UNITS: **mm**	
DRAWING No.: **VC-103**	REV: **A**	SHEET: **4 / 7**	
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES / DATE: 2026-09-09